Zero-inflated models

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Physician visits

Data on 4406 individuals, age 66 and over, who are covered by Medicare. Variables include:

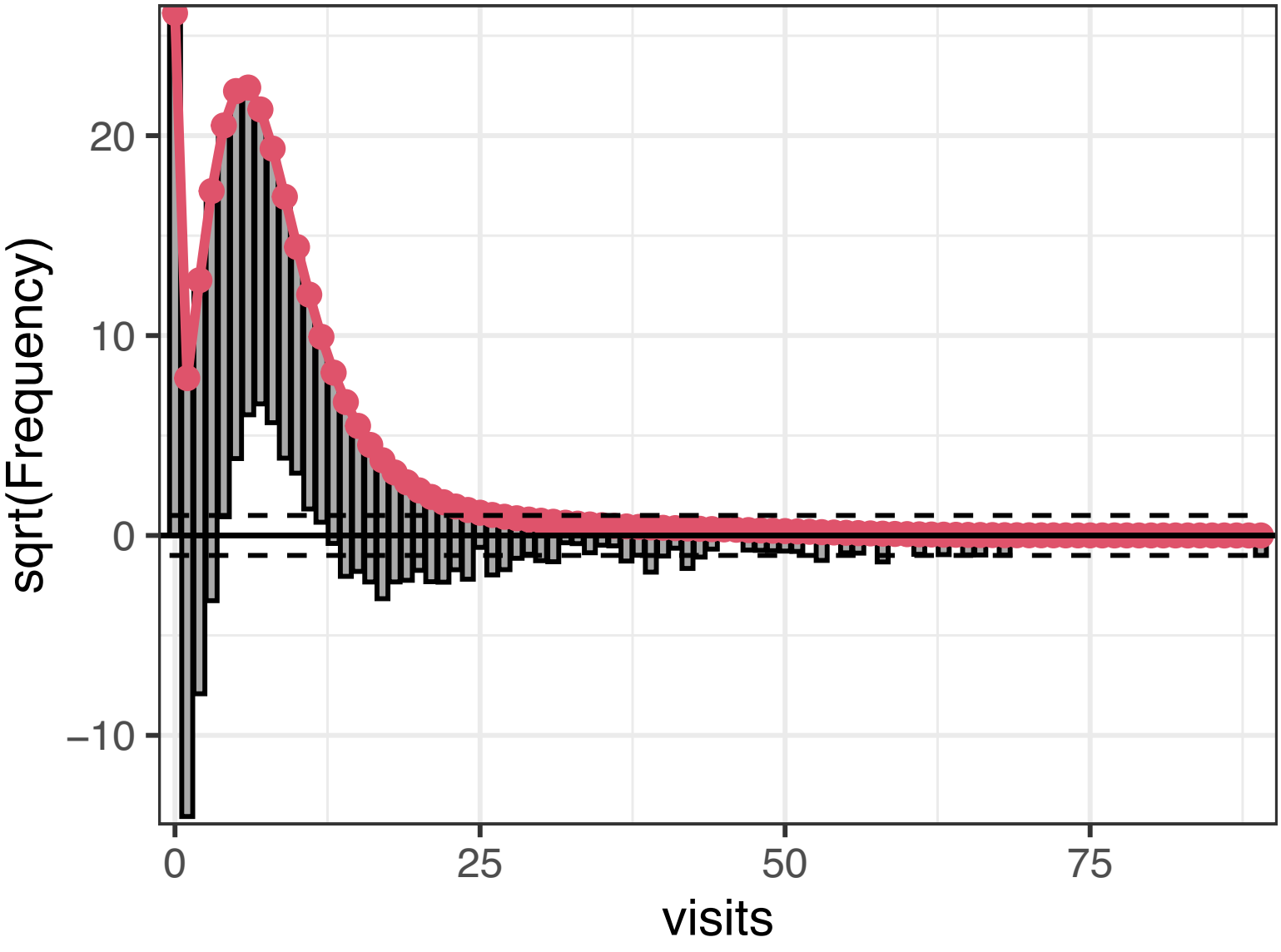
- `visits`: Number of physician office visits
- `hospital`: Number of hospital stays
- `health`: Factor indicating self-perceived health status, levels are "poor", "average", "excellent"
- `chronic`: Number of chronic conditions
- `gender`: Factor indicating gender
- `school`: Number of years of education
- `insurance`: Factor. Is the individual covered by private insurance?

Goal: Model the number of physician visits.

Activity 1

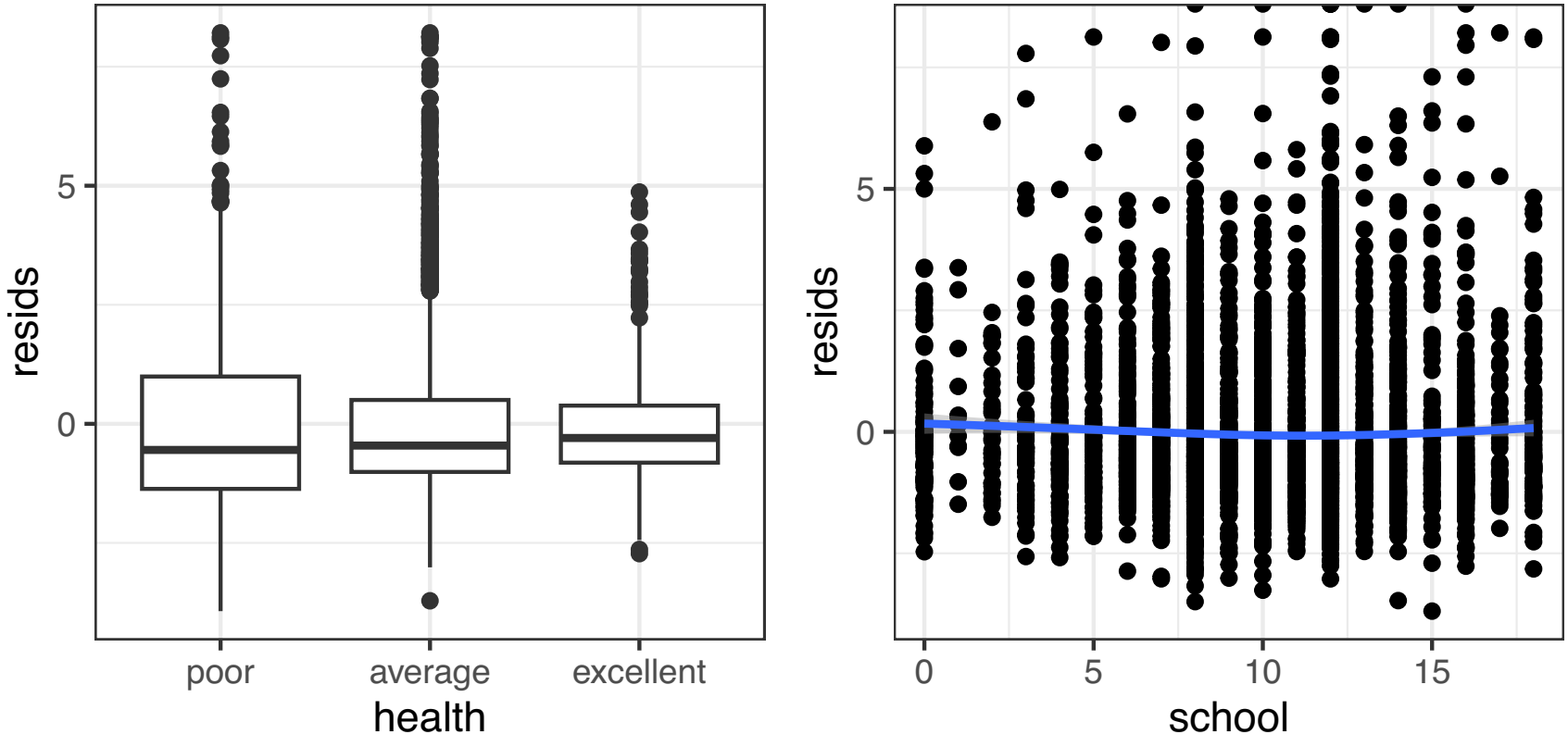
Work on the class activity (course website).

ZIP diagnostics



ZIP diagnostics

- right-skewed residuals
- more variability in residuals than $N(0,1)$
- shape looks pretty good



How would you modify the model?

"What can I do differently?"

A guiding principle for the course.

- Linear regression is a poor choice for some response variables
  - Modification: GLMs!
- Not all count variables are Poisson
  - Modifications: sandwich variance, quasi-Poisson, negative binomial
- There are more 0s than a Poisson distribution accounts for
  - Modification: ZIP
- There are still excess 0s, but variance is still too high for a Poisson
  - Modification: zero-inflated negative binomial (ZINB)

Zero-inflated count regression

Latent variable: $Z_i \sim \text{Bernoulli}(p_i)$

$Y_i \mid Z_i = 1, x_i = 0$

$Y_i \mid Z_i = 0, x_i \sim \underbrace{f(y; \xi_i)}$

pmf of some count distribution with parameters $\xi_i = (\mu_i, \dots)$

$\uparrow E[Y_i \mid Z_i = 0, x_i]$

(common choices: Poisson, NB)

Poisson: $\xi_i = \mu_i$

NB: $\xi_i = (\mu_i, r)$ ← size parameter

$$P(Y_i = y \mid x_i) = \begin{cases} p_i + (1-p_i) f(0; \xi_i) & y = 0 \\ (1-p_i) f(y; \xi_i) & y > 0 \end{cases}$$

$$\log(\mu_i) = x_i^T \beta$$

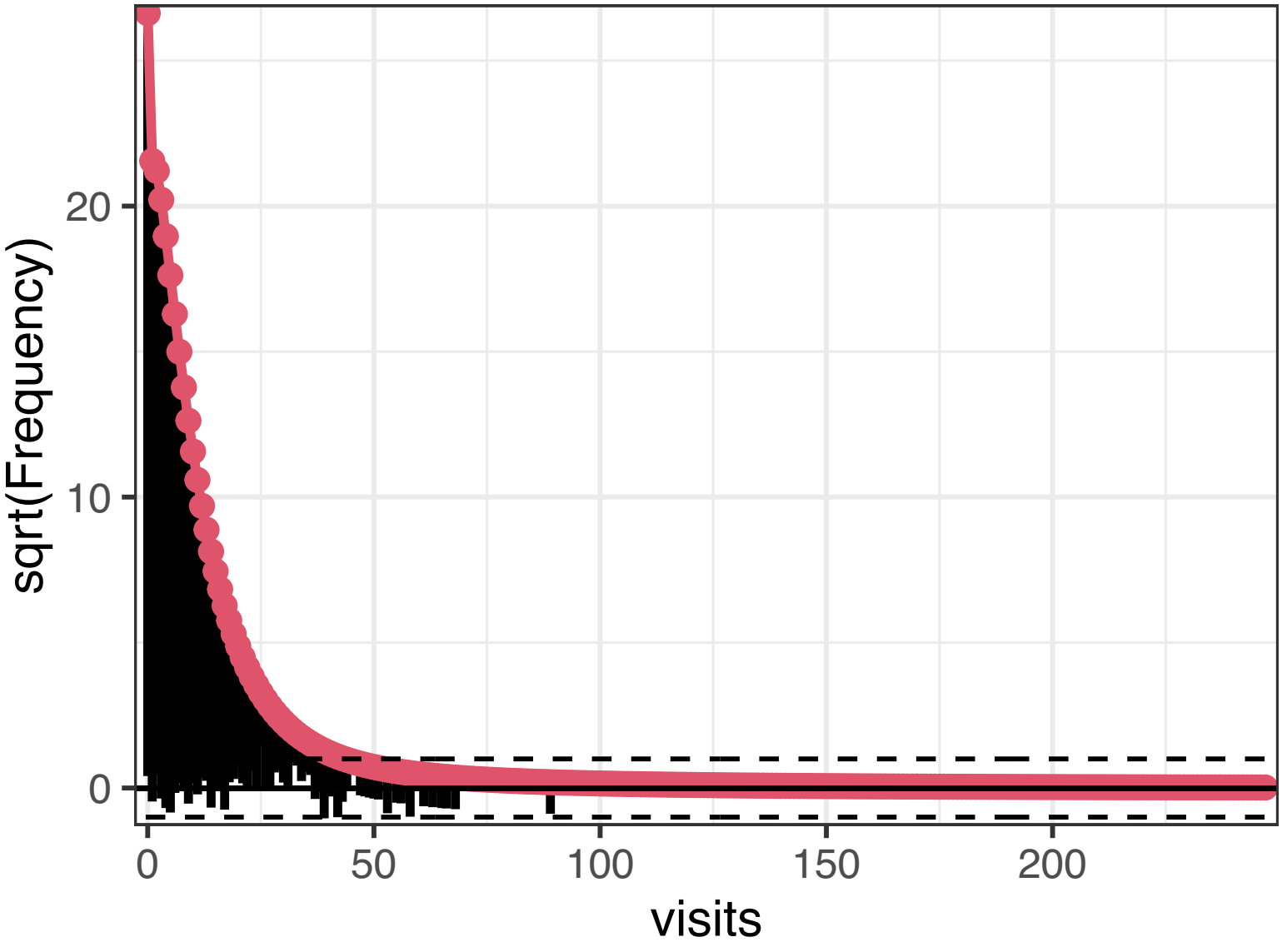
(possible to use a different link)

$$\log\left(\frac{p_i}{1-p_i}\right) = x_i^T \gamma$$

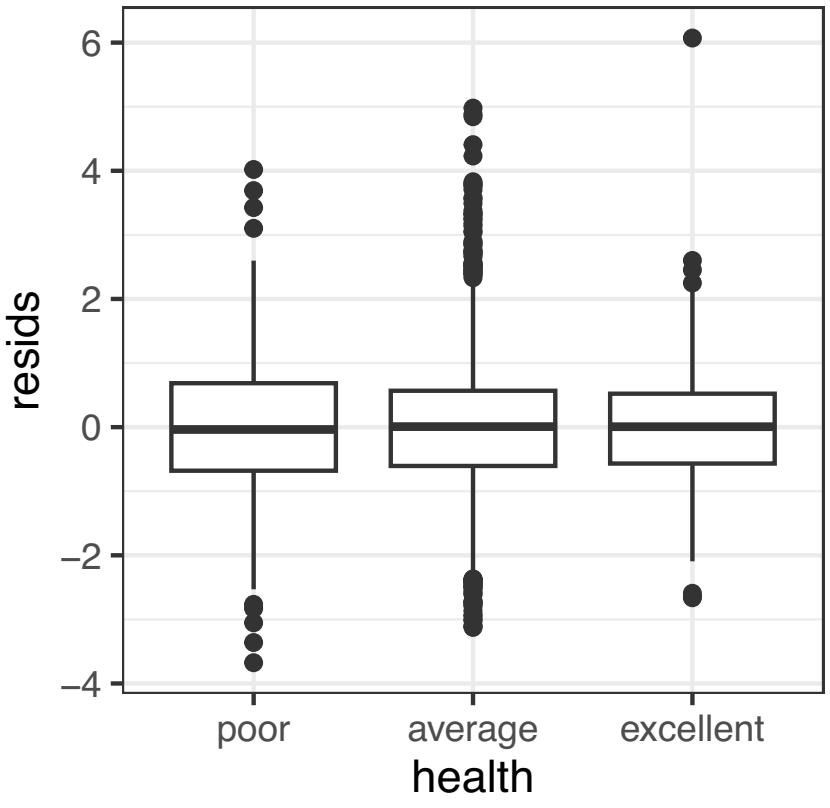
Activity 2

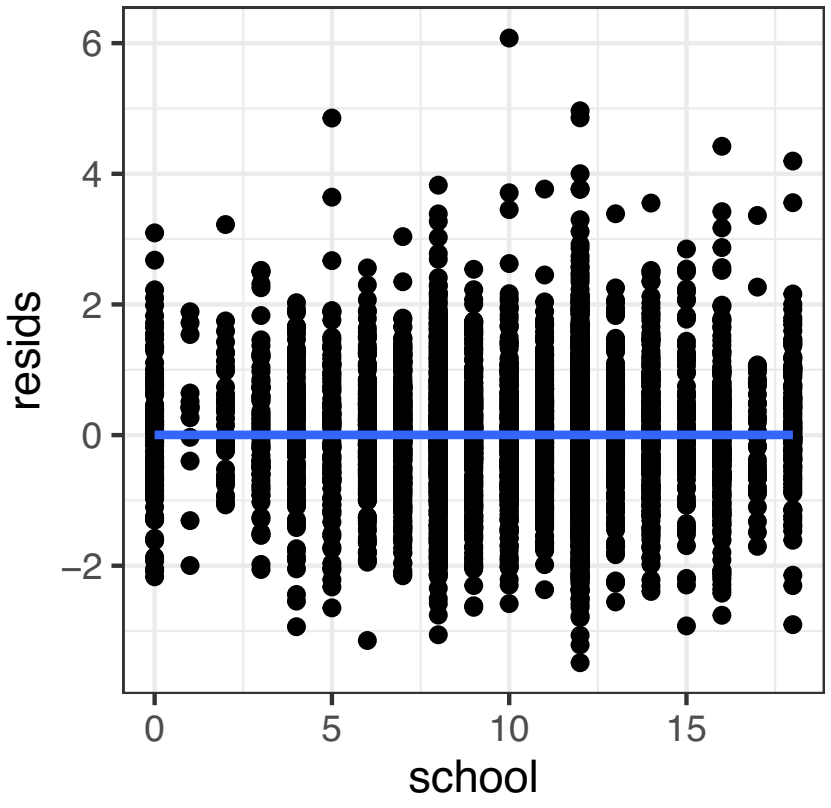
Work on the class activity (course website).

ZINB diagnostics



ZINB diagnostics





Inference

We are interested in whether there is any relationship between an individual's self-perceived health status (`health`) and their number of physician visits (`visits`), after accounting for other explanatory variables.

Question: How would I carry out a hypothesis test to address this question?

Inference

```
m2 <- pscl::zeroinfl(visits ~ .,
                     dist = "negbin", data = nmes)
m3 <- pscl::zeroinfl(visits ~ . - health,
                     dist = "negbin", data = nmes)

2*(m2$loglik - m3$loglik)
```

```
## [1] 70.85807
```

```
pchisq(2*(m2$loglik - m3$loglik), df=4, lower.tail=F)
```

```
## [1] 1.495596e-14
```